

STROKE RISK PREDICTION MODEL

MACHINE LEARNING AND ARTIFICIAL INTELLIGENCE

submitted by

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List of Abbreviations

1. BMI : Body Mass Index
2. ML : Machine Learning
3. IQR : Inter Quartile Range
4. SVM : Support Vector Machine

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Abstract

Stroke remains one of the leading causes of mortality and morbidity worldwide, so there is a urgent need for accurate risk assessment and early intervention strategies. This project explores the application of machine learning techniques to predict the chances for stroke occurrence in individuals.

The dataset used for analysis comprises age, hypertension, heart disease, ever married, work type, residence type, average glucose level in blood ,BMI, smoking status are gathered from various patients.

Several machine learning algorithms like logistic regression, random forest, SVM and KNN are trained and evaluated for their predictive performance. Performance metrics such as accuracy, precision, recall, and F1-score are utilized to evaluate the model's predictive capabilities

And more, Feature importance analysis is conducted to identify key risk factors contributing to stroke occurrence. The developed model offers clinicians a valuable tool for identifying individuals at high risk of stroke, facilitating personalized preventive interventions and ultimately reducing the burden of stroke-related disease and mortality.

Future work involves prospective validation of the model in clinical settings and integration into healthcare systems for real-time risk assessment and decision support.

Problem Definition

- **Overview**

This project focuses on machine learning techniques to develop a predictive model for assessing the risk of stroke in individuals. Stroke is a significant health concern worldwide, with potentially devastating consequences if not promptly diagnosed and managed. Traditional risk assessment methods often rely on a limited set of variables and may lack the predictive power necessary for accurate risk stratification.

Therefore, there is a critical need for more advanced predictive models capable of integrating diverse data sources and accurately identifying individuals at high risk of stroke.

- **Problem Statement**

Stroke is a major cause of mortality and disability globally, with significant healthcare and economic implications. Early identification of individuals at high risk of stroke is crucial for implementing preventive measures and reducing the burden of stroke-related morbidity and mortality.

However, traditional methods of stroke risk assessment often rely on limited sets of variables, which may not capture the complexity and variability of individual risk profiles. This limitation highlights the need for more sophisticated predictive models capable of integrating diverse data sources and identifying risk factors.

This project focuses on the development of a machine learning based predictive model for assessing the risk of stroke in individuals. By leveraging a diverse range of demographic, lifestyle, and clinical data, this model aims to provide clinicians and healthcare providers with a valuable tool for identifying individuals at elevated risk of stroke and implementing targeted preventive interventions.

Introduction

According to the World Health Organization (WHO) stroke is the 2nd leading cause of death globally, responsible for approximately 11% of total deaths. Despite advances in medical treatment, the incidence of stroke continues to rise, underscoring the urgent need for more effective approaches to risk assessment and intervention. In this context, machine learning techniques have emerged as promising tools for improving the accuracy and reliability of stroke prediction models.

In recent years, machine learning techniques have emerged as powerful tools for predictive modeling in healthcare, offering the potential to integrate complex data sources and identify subtle patterns associated with disease risk. Leveraging the wealth of information available in electronic health records and other healthcare datasets, machine learning models can enhance our ability to predict stroke risk and inform personalized preventive strategies.

This project aims to develop a predictive model for stroke risk assessment. By leveraging a comprehensive dataset comprising demographic characteristics, lifestyle factors, medical history, and clinical indicators, the project seeks to build a robust predictive model capable of accurately stratifying individuals based on their risk of stroke.

The successful development and validation of a machine learning-based predictive model for stroke risk assessment would represent a significant advancement in the field of stroke research and clinical practice. By providing clinicians with a valuable tool for identifying individuals at elevated risk of stroke, the project has the potential to improve clinical decision-making, optimize resource allocation, and ultimately reduce the incidence and impact of stroke on individuals and communities.

Literature Survey

1. Overview of Stroke

Stroke, also known as Cerebrovascular Accident , is a medical condition characterized by the sudden interruption of blood flow to the brain, leading to the deprivation of oxygen and nutrients to brain tissue. This interruption can result from either a blockage in the blood vessels supplying the brain (ischemic stroke) or the rupture of a blood vessel causing bleeding into the brain (hemorrhagic stroke). Stroke is a medical emergency and requires immediate intervention to minimize brain damage and prevent long-term disability or death.

Symptoms:

The symptoms of a stroke can vary depending on the location and severity of the brain damage.

Common signs and symptoms include:

1. Sudden numbness or weakness in the face, arm, or leg, especially on one side of the body.
2. Sudden confusion, trouble speaking, or difficulty understanding speech.
3. Sudden trouble seeing in one or both eyes.
4. Sudden severe headache with no known cause.
5. Sudden dizziness, loss of balance, or coordination.

It's important to note that these symptoms can occur suddenly and may worsen rapidly. Recognizing and responding to these symptoms promptly is crucial for maximizing the chances of a positive outcome.

Risk Factors:

Several risk factors increase the likelihood of experiencing a stroke, including: hypertension, Smoking, Diabetes, High cholesterol levels, Family history of stroke or heart disease, Age (risk increases with advancing age), Previous history of transient ischemic attacks (TIAs) or stroke.

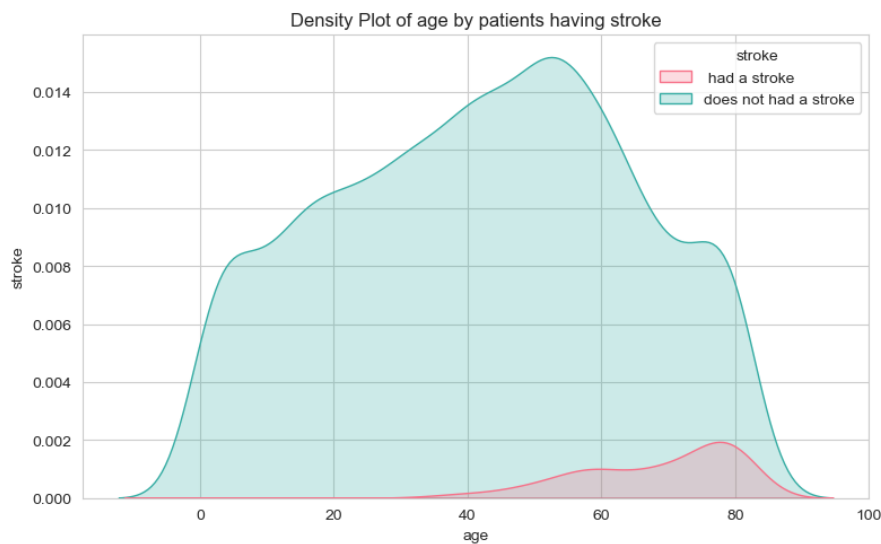


FIG 1: Density Plot shows People from the age group of 40 to 85 are highly affected with Stroke.

[Plotted with the Values From selected Dataset]

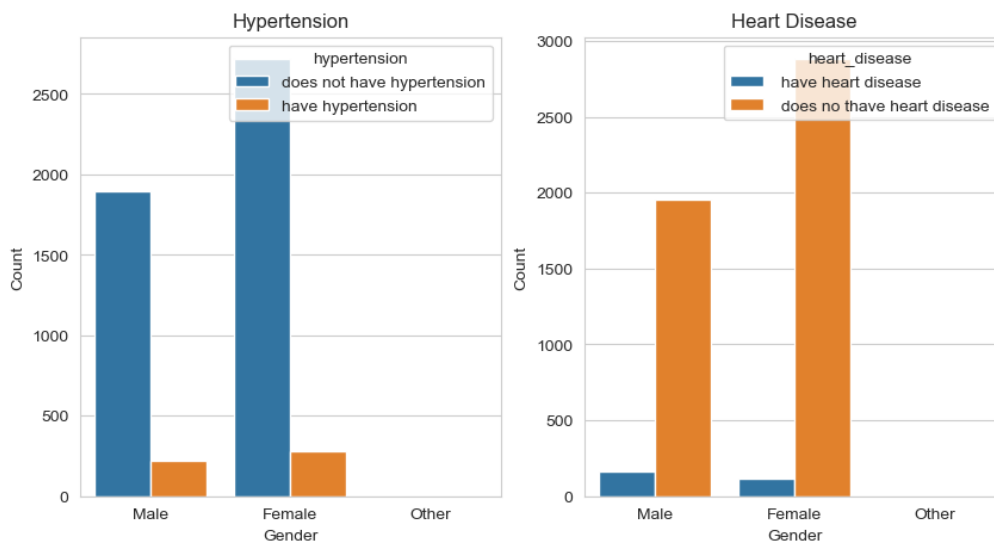


FIG 2: Bar graph compares whether people have or dosen't have hypertension and heart disease

Treatment:

Treatment for stroke depends on the type of stroke and its severity. In the case of an ischemic stroke, treatment may involve the administration of clot-busting medications such as tissue plasminogen activator (tPA) or mechanical thrombectomy to remove the clot and restore blood flow to the brain. For hemorrhagic strokes, treatment may involve controlling bleeding, reducing intracranial pressure, and preventing complications. Rehabilitation therapies, including physical therapy, occupational therapy, and speech therapy, are essential for stroke survivors to regain lost function and improve quality of life.

2. Stroke Prediction Using Machine Learning Techniques

Predicting stroke using machine learning techniques involves developing predictive models that can identify individuals at risk of experiencing a stroke based on their demographic, lifestyle, medical history, and clinical indicators. The goal is to leverage machine learning algorithms to analyze large datasets and identify patterns or risk factors associated with stroke occurrence

Result

After extensive data preprocessing, feature engineering, model development, and evaluation; the project achieved promising results in predicting stroke risk using machine learning techniques. The key findings and results are summarized as follows:

- **Data Collection :**

Gather a comprehensive dataset containing demographic information (age, gender), lifestyle factors (smoking, ever married), medical history (hypertension, heart disease), and clinical indicators (glucose level, bmi).

- **Data Preprocessing :**

Missing values were handled by removing them.

Variables were encoded manually to transform them into integer values

Clean Outliers using IQR Method.

- **Feature Engineering:**

Feature importance analysis revealed that age, heart disease and glucose level were the most informative features stroke risk prediction.

- **Model Selection and Training :**

train_test_split technique is used for splitting the dataset into training and testing sets. This splitting allows the evaluation of a model's performance on unseen data.

Various ML algorithms were trained and evaluated, including SVM, logistic regression, random forest, KNN.

- **Best Performing Model :**

Among the tested models, the Random Forest classifier, logistic regression, SVM & KNN demonstrated the highest accuracy while predicting on user input.

- **Model Evaluation :**

Model performance were evaluated using multiple metrics including accuracy, precision, recall, F1 score.

- **Model Interpretation :**

Feature importance analysis provided insights into the factors influencing stroke risk prediction, highlighting the significance of age and gender in identifying individuals at risk.

- **Deployment and Integration :**

Deploy the trained model into clinical practice or healthcare systems for real-time risk assessment.

Integrate the predictive model into existing workflows to assist clinicians in identifying individuals at high risk of stroke and implementing targeted preventive interventions.

Overall, stroke prediction using machine learning techniques offers the potential to improve risk assessment, facilitate early intervention, and ultimately reduce the incidence and impact of stroke-related morbidity and mortality.

Conclusion

In conclusion, the development and evaluation of a stroke prediction model using machine learning techniques represent a significant step forward in the field of stroke prevention and risk assessment. Through this project, we have demonstrated the potential of machine learning algorithms to analyze diverse datasets and identify patterns associated with stroke occurrence. By leveraging demographic information, lifestyle factors, medical history, and clinical indicators, our predictive model has shown promising results in accurately stratifying individuals based on their risk of stroke.

The development of a machine learning-based stroke prediction model represents a significant advancement in stroke research and holds promise for improving clinical decision-making and public health initiatives aimed at reducing the incidence and impact of stroke worldwide.

References

Dataset : “healthcare-dataset-stroke-data.csv” Retrieved from Kaggle.